

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION
January, 2018

Probability (Ph.D. Version)

Instructions to the Student

- a. Answer all six questions. Each will be graded from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.
 - c. Keep scratch work on separate pages in the same booklet.
 - d. If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.
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1. Let f be a continuous bounded function on $\mathbb{R}$. Compute (with justification)

$$\lim_{n \rightarrow \infty} \int_0^\infty \dots \int_0^\infty f\left(\frac{x_1 + \dots + x_n}{n}\right) e^{-(x_1 + \dots + x_n)} dx_1 \dots dx_n.$$

2. Let $\xi_1, \xi_2, \dots$ be independent exponential random variables with parameter 1. Prove that

$$\limsup_{n \rightarrow \infty} \frac{\xi_n}{\ln n} = 1.$$

3. Suppose that ξ has the following properties:

- (a) $\xi \geq 0$.
- (b) $E\xi = 6$.
- (c) $\text{Var}\xi = 6$.

By using (a)-(c), give an estimate

$$P(\xi \geq 10) \leq r$$

with the constant r that you should choose to be as small as you can.

4. Consider a time-homogeneous Markov chain X_n , $n \geq 0$, on the state space $\mathbb{N}$. Suppose that

$$P(X_{n+1} = m - 1 | X_n = m) = P(X_{n+1} = m + 1 | X_n = m) = 1/2 \text{ if } m > 1, \text{ } m \text{ is odd,}$$

$$P(X_{n+1} = m - 1 | X_n = m) = 2/3, \quad P(X_{n+1} = m + 1 | X_n = m) = 1/3 \text{ if } m \text{ is even.}$$

Prove that

$$P(\inf_{n \geq 0} X_n = 1) = 1.$$

5. Let W_t be a Brownian motion. Calculate $E(W_t^2 | W_s)$, $E(W_s | W_t)$, and $E(W_s^2 | W_t)$ for $0 \leq s \leq t$.

Hint: you can write $W_s = \frac{s}{t}W_t + (W_s - \frac{s}{t}W_t)$.

6. Does there exist a Borel-measurable set $A \subseteq [0, 1]$ such that

$$\lambda(A \cap [a, b]) = \frac{1}{2}(b - a)$$

for each interval $[a, b] \subseteq [0, 1]$, where λ is the Lebesgue measure on $[0, 1]$.